77. Match List I (Fuel Gases) with List II (Major Constituents) and select the correct answer using the codes given below the lists:

List I		List II
A. CNG	1.	Carbon Monoxide, Hydrogen
B. Coal gas	2.	Butane, Propane
C. LPG	3.	Methane, Ethane
D. Water gas	4.	Hydrogen, Methane, Carbon Monoxide

	A	B	C	D
(a)	2	1	3	4
(b)	3	4	2	1
(c)	2	4	3	1
(d)	3	1	2	4

Ans. (b) : Compressed Natural Gas (CNG) is made by compressing natural gas which is composed of methane (CH_4), it also contain small amount of ethane. Coal gas typically contains hydrogen, methane and carbon monoxide. LPG is the abbreviation of Liquefied Petroleum Gas. The major constituents of LPG are propane and butane. Water gas is a synthesis gas, containing carbon monoxide and hydrogen.

78. Standard 18-carat gold sold in the market contains

(a) 82 parts gold and 18 parts other metals.
(b) 18 parts gold and 82 parts other metals.
(c) 18 parts gold and 6 parts other metals.
(d) 9 parts gold and 15 parts other metals.

Ans. (c) : 24 carat is pure gold with no other metals. Lower cartages contain less gold; 18 carat gold contains 75 per cent gold and 25 per cent other metals, often copper or silver.

79. Salts of which of the following elements provide colours to fireworks?

(a) Zinc and sulphur
(b) Potassium and mercury
(c) Strontium and barium
(d) Chromium and nickel

Ans. (c) : Creating firework colors is a complex endeavor, requiring considerable art and application of physical science. Strontium and barium both are alkaline earth metal and are extremely reactive. They both impart characteristic color to flame. Strontium salts impart a red color to fireworks. Strontium compounds are also important for stabilizing fireworks mixtures. Barium is used to create green colors in fireworks, and it can also help stabilize other volatile elements.

BIOLOGY

80. Consider the following statements:

1. **Non-function of Lacrimal gland is an important symptom deficiency of Vitamin A.**
2. **Deficiency of Vitamin B1 can lead to indigestion and heart enlargement.**
3. **Vitamin C deficiency can lead to pain in the muscles.**
4. **Deficiency of Vitamin D causes increased loss of Ca^{++} in urine.**

Which of the statements given above are correct?

(a) 1 and 2 (b) 2, 3 and 4
(c) 1, 3 and 4 (d) 1, 2, 3 and 4

Ans. (d) : Lacrimal is a tear producing gland which does not function due to deficiency of vitamin -A. Deficiency of vitamin - B1 causes loss of appetite, heart disorders, muscular weakness etc.

81. The hormone insulin is a

(a) Glycolipid (b) Fatty acid
(c) Peptide (d) Sterol

Ans. (c) : Insulin is a peptide hormone composed of 51 amino acids. Insulin is secreted from pancreas (Islets of Langerhans).

82. Consider the following statements:

1. **Toothless mammals such as pangolins are not found in India.**
2. **Gibbon is the only ape found in India.**

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b) : Pangolins (Manis crassicaudata) are found in India, Sri Lanka, Nepal, Africa and some parts of Pakistan. Gibbons are the only apes found in India. The Hoollongapar Gibbon Sanctuary is in Assam which is a safe recluse for gibbons.

83. In which organ of the human body are the lymphocyte cells formed?

(a) Liver (b) Long bone
(c) Pancreas (d) Spleen

Ans. (b) : Lymphocytes are the cells that determine the specificity of the immune response to infectious microorganisms and other foreign substances. In humans, lymphocytes make up 25 to 33% of the total number of leukocytes. They are found in central lymphoid organs and tissues such as the spleen, tonsils and lymph nodes and large bones (Bone Marrow).

84. Assertion (A) : Fatty acids should be a part of the balanced human diet.

Reason (R) : The cells of the human body cannot synthesize any fatty acids.

(a) Both A and R are individually true and R is the correct explanation of A.
(b) Both A and R are individually true but R is NOT the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

Ans. (c) : Assertion is true because a balance diet is a diet which contains the correct amount of carbohydrates, proteins and fats. But the reason is false, because human body cells synthesize various fatty acids except linoleic and linolenic fatty acids.

85. Consider the following statements:

1. Femur is the longest bone in the human body.
2. Cholera is a disease caused by bacteria.
3. 'Athlete's foot' is a disease caused by virus.

Which of the statements given above are correct?

(a) 1 and 2 (b) 2 and 3
(c) 1 and 3 (d) 1, 2 and 3

Ans. (a) : The head of the femur articulates with the acetabulum in the pelvic bone forming the hip joint, while the distal part of the femur articulates with the tibia and kneecap forming the knee joint. By most measures the femur is the strongest bone in the body. The femur is also the longest bone in the human body. Cholera is caused by a bacteria Vibrio cholerae.Athlete's foot disease is caused by parasific fungus of genus Trichophyton.

ENVIRONMENT & ECOLOGY

86. Match List I (National Park/Sanctuary) with List II and (State) and select the correct answer using the codes given below the lists:

List I		List II
A. Kanger Ghati National Park	1.	Chhattisgarh
B. Nagarahole National Park	2.	Haryana
C. Kugti Wildlife Sanctuary	3.	Himachal Pradesh
D. Sultanpur Bird Sanctuary	4.	Karnataka

	A	B	C	D
(a)	3	2	1	4
(b)	1	4	3	2
(c)	3	4	1	2
(d)	1	2	3	4

Ans. (b) : National Park/ Sanctuary

A.	Kanger Ghati National Park	- Chhattisgarh
B.	Nagarahole National Park	- Karnataka
C.	Kugti Wildlife Sanctuary	- Himachal Pradesh
D.	Sultanpur Bird Sanctuary	- Haryana

87. Match List I (Biosphere Reserve) with List II (States) and select the correct answer using the codes given below the lists:

List I		List II
A. Similipal	1.	Sikkim
B. Dihang Dibang	2.	Uttaranchal
C. Nokrek	3.	Arunachal Pradesh
D. Kanchenjunga	4.	Orissa
	5.	Meghalaya

	A	B	C	D
(a)	1	3	5	4
(b)	4	5	2	1
(c)	1	5	2	4
(d)	4	3	5	1

Ans. (d) : Similipal biosphere reserve is in Orissa. Dihang Dibang biosphere reserve is in Arunachal Pradesh, Nokrek biosphere reserve is in Meghalaya and Kanchenjunga biosphere reserve is situated in Sikkim.

88. Amongst the following Indian States which one has the minimum total forest cover?

(a) Sikkim (b) Goa
(c) Haryana (d) Kerala

Ans. (c) : Haryana has the minimum total forest cover (till date 6.79%).

CURRENT AFFAIRS

89. Which of the following is not a recommendation of the task force on direct taxes under the chairmanship of Dr. Vijay L. Kelkar in the year 2002?

(a) Abolition of Wealth Tax.
(b) Increase in the exemption limit of personal income to Rs. 1.20 lakh for widows.
(c) Elimination of standard deduction.
(d) Exemption from tax on dividends and capital gains from the listed equity.